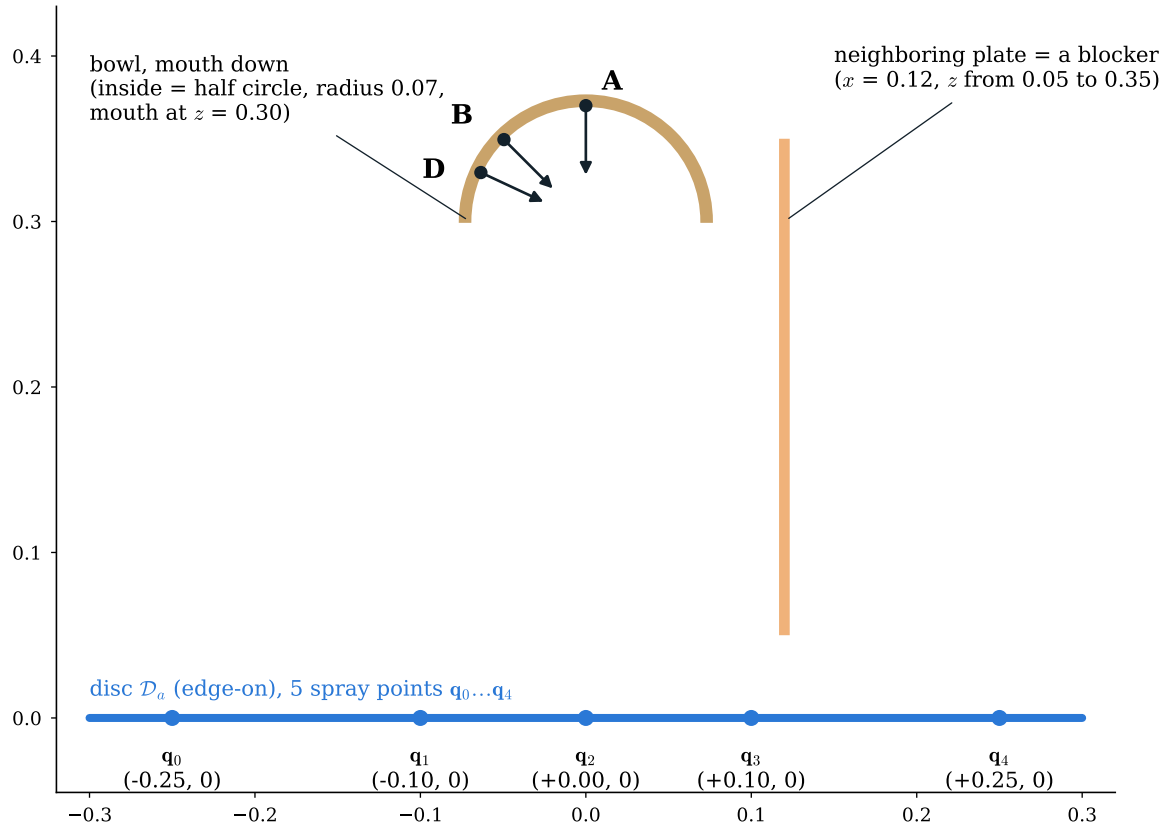


Toy setup (side view, meters, to scale)



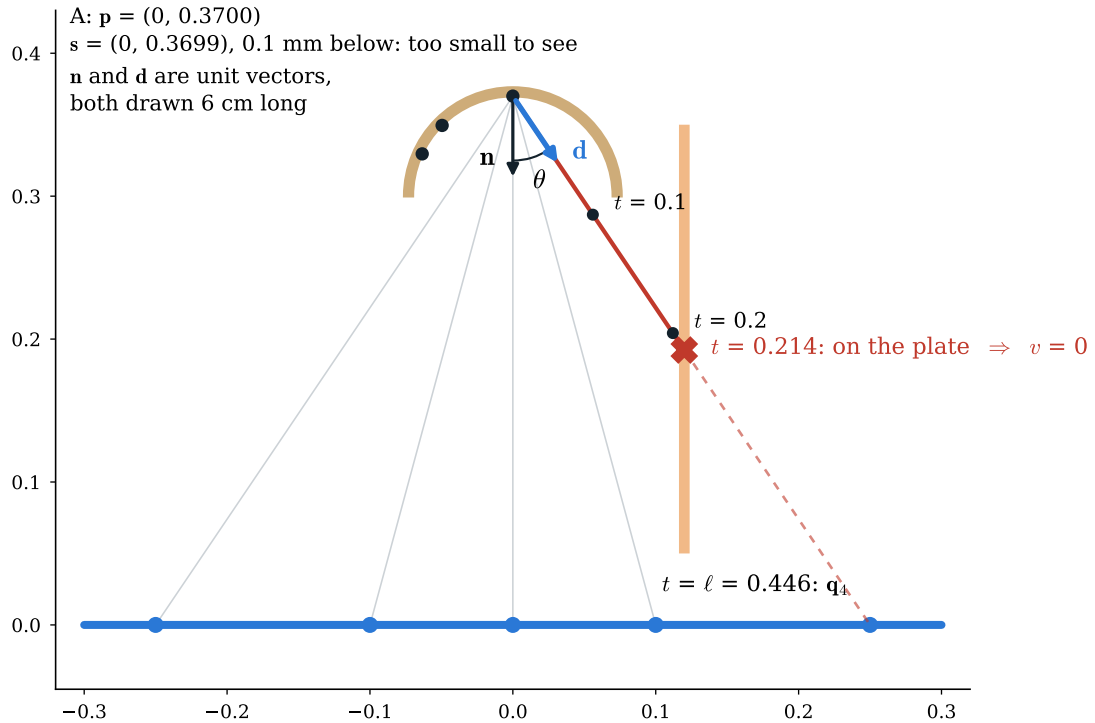
Three dots on the bowl's inside, each with its unit normal $\mathbf{n}$ (arrow, into the cavity):

A: $\mathbf{p} = (+0.0000, 0.3700)$, $\mathbf{n} = (+0.000, -1.000)$

B: $\mathbf{p} = (-0.0495, 0.3495)$, $\mathbf{n} = (+0.707, -0.707)$

D: $\mathbf{p} = (-0.0634, 0.3296)$, $\mathbf{n} = (+0.906, -0.423)$

One line, step by step: dot A → spray point q_4

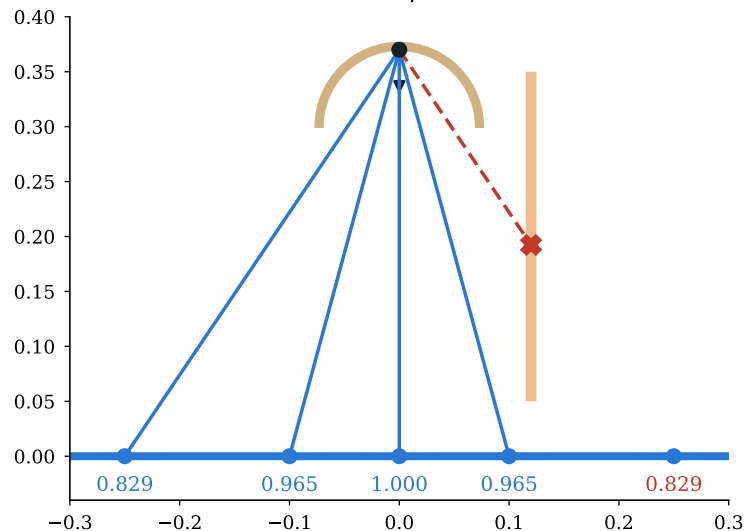


1. $\mathbf{n} = (0, -1)$: the top of a mouth-down bowl faces straight down
2. $\mathbf{p} = (0, 0.3700)$
3. $\mathbf{s} = \mathbf{p} + \epsilon \mathbf{n} = (0, 0.3700) + 0.0001 \cdot (0, -1) = (0, 0.3699)$
4. $\mathbf{q}_4 - \mathbf{s} = (0.25, -0.3699)$, $\ell = \sqrt{0.25^2 + 0.3699^2} = 0.446$ m
5. $\mathbf{d} = (\mathbf{q}_4 - \mathbf{s}) / \ell = (0.560, -0.829)$ (length 1)
6. walk $\mathbf{s} + t\mathbf{d}$: $t = 0$ is $\mathbf{s}$, $t = 0.446$ is $\mathbf{q}_4$; at $t = 0.214$ the point is $(0.120, 0.192)$, on the plate
7. so the set {points of the line} $\cap$ {blockers} is not empty $\Rightarrow v = 0$ (blocked)
8. $\gamma = \max(0, \mathbf{n}^\top \mathbf{d}) = 0 \cdot 0.560 + (-1)(-0.829) = 0.829 = \cos \theta$

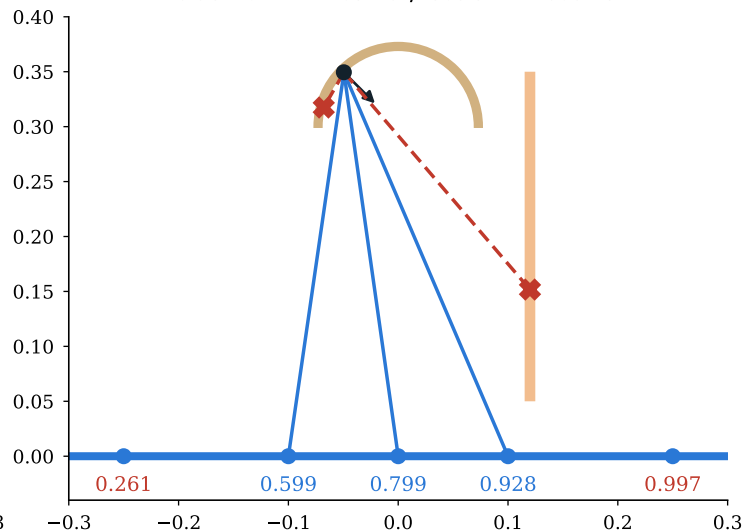
All 15 lines: 3 dots $\times$ 5 spray points

— open: $v = 1$ ✖ blocked (plate or own wall): $v = 0$ behind the tangent plane: $\gamma = 0$ (ignored)

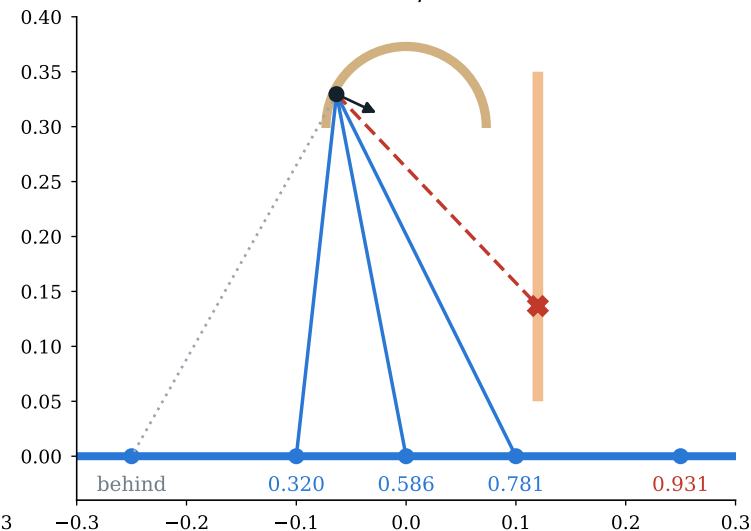
dot A: $e = 3.759 / 4.588 = 0.819$



dot B: $e = 2.326 / 3.584 = 0.649$



dot D: $e = 1.687 / 2.618 = 0.644$



numbers under the disc points: γ of each line. $e = (\text{sum of } \gamma \text{ over open lines}) / (\text{sum of } \gamma \text{ over all front-facing lines})$

If these 3 dots were the whole bowl (Step 4): $E = (0.819 + 0.649 + 0.644) / 3 = 0.704$